

Yesmine Ayeb

Agronomy Engineer | MSc in Horticultural Genetics & Biotechnology

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EDUCATION

- **CIHEAM - Mediterranean Agronomic Institute of Chania** 2024 - 2026
MSc in Horticultural Genetics and Biotechnology Chania, Greece
- **Higher Institute of Agronomic Sciences of Chott Mariem** 2019 - 2022
Agronomy Engineering - Speciality in Horticultural Production Sousse, Tunisia
- **Agrocampus Ouest Rennes-Angers** 2021 - 2022
Erasmus Exchange Program Angers, France
- **Bizerte Preparatory Engineering Institute** 2017 - 2019
Pre-Engineering courses in Biology - Geology Bizerte, Tunisia

RESEARCH EXPERIENCE

- **Plant Molecular Biology and Biotechnology Laboratory, CIHEAM MAICH** Oct 2025 - Jun 2026
MSc Thesis: Functional Characterization of Greek Tomato Cultivars Chania, Greece
 - Characterized seven Greek local tomato cultivars and two reference cultivars for ripening progression, ethylene production by GC, firmness, water loss, and postharvest marketability.
 - Integrated physiological phenotypes with red-ripe RNA-seq data and analysed cultivar-dependent differences in R.
Additional Research: Generation and Characterization of Tomato SIP4H Transgenic Lines Chania, Greece
 - Assembled CRISPR/Cas9 knockout constructs targeting *SIP4H3* and *SIP4H6* using GoldenBraid, including *E. coli* transformation, blue-white screening, plasmid miniprep, PCR and gel validation.
 - Performed *Agrobacterium*-mediated tomato transformation and tissue culture, followed by PCR and TaqMan qPCR screening of regenerated transgenic lines.
 - Applied HypoxyLab low-oxygen treatments followed by RNA extraction, cDNA synthesis and SYBR Green RT-qPCR.
 - Performed protein extraction, SDS-PAGE and Western blotting.
- **Horticultural Science Laboratory, National Agronomic Institute of Tunisia** Jun 2022 - Dec 2022
Final-Year Research Project: Tissue Culture of Rosa centifolia Tunis, Tunisia
 - Optimised *Rosa centifolia* micropropagation initiation by evaluating IBA concentrations to improve explant establishment and reduce oxidative browning.
 - Evaluated rosemary and thyme essential oils and a microalgal consortium as antimicrobial treatments against conventional surface disinfection.

ADDITIONAL EXPERIENCE

- **Azib Flower** Jun 2021
Internship : Ornamental greenhouse production and Integrated Pest Management (IPM) Bizerte, Tunisia
- **Sanlucar** Aug 2020
Internship : Hydroponic red berry production, flowering analysis, water and nutrient management Bizerte, Tunisia

SKILLS

- **Molecular Biology:** DNA/RNA/Protein extraction | PCR | RT-qPCR | Gel Electrophoresis | Western Blot | SDS-PAGE | CRISPR/Cas9 | Molecular Cloning (GoldenBraid)
- **Plant Biotechnology:** Tissue Culture | *Agrobacterium*-mediated Transformation | HypoxyLab
- **Analytical:** HS-SPME-GC-MS
- **Programming & Software:** R | ImageJ | RStudio | Positron | Overleaf
- **Languages:** Arabic: Maternal | English: C1 (Academic IELTS 7.5) | French: B2
- **Other:** Driving licence B | First Aid | Fire Safety